



through Artificial Intelligence
Tools|Effective Analysis of Genetic Diversity of Plant Species
through Artificial Intelligence
Tools

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Abstract:

Genetic variation within species arises from complex interactions between evolutionary processes and environmental factors, often reflected in polymorphic patterns of Single-Nucleotide Polymorphisms (SNPs). Traditional landscape genetics approaches have primarily focused on either broad-scale or local-scale environmental effects, limiting the statistical inference of genetic diversity. In this study, we present a comprehensive statistical framework integrating supervised machine learning, convolutional neural networks (CNNs), and simulation-based modeling to quantify the influence of multiple factors on genetic variation. Supervised machine learning allows prediction of key genetic metrics, such as allelic richness and population differentiation, while simultaneously estimating the relative importance of environmental and historical drivers. CNNs process SNP alignments as high-dimensional data, extracting patterns without requiring predefined summary statistics or population assignments. Simulation-based approaches, including Approximate Bayesian Computation (ABC), enable robust estimation of parameter posteriors under complex evolutionary scenarios. By combining these methods, our framework captures both discrete and continuous genetic processes, models stochasticity in natural populations, and provides interpretable statistical measures of factor importance. This approach enhances inference across spatial and temporal scales, facilitating hypothesis testing and evolutionary interpretation. The framework also has practical applications for conservation planning and biodiversity management, allowing researchers to identify populations and environmental factors most critical for maintaining genetic diversity. This integrative, statistics-driven approach represents a significant advancement in molecular ecology, offering scalable, interpretable, and robust tools for analyzing genomic variation in natural populations.

Keywords: Supervised machine learning, Convolutional neural networks, Statistical genomics

Mathematics Subject Classification (2010): 62P10, 62F15, 92D20, 92D15.

1 Introduction

Quantitative understanding of genetic variation in spatial and environmental contexts is a central objective of landscape genetics. Historically, molecular studies used organellar genes or allelic markers such as microsatellites to quantify variation, while sequence-based data informed deeper evolutionary inference (Holderegger & Wagner, 2008). Single-Nucleotide Polymorphisms (SNP) data has revolutionized statistical genetics by providing dense, genome-wide markers that enable hypothesis testing at multiple scales (Eaton & Overcast, 2020). Despite these advances, selecting appropriate analytical methods remains a key statistical challenge. Classical population genetics models often assume discrete populations, migration-drift equilibrium, and neutrality, which may not hold in complex landscapes. Mis-specification of populations or unrecognized continuous processes can bias parameter estimation and reduce statistical power. Similarly, approximate Bayesian computation (ABC) and related model-based approaches require careful selection of summary statistics, which can influence posterior estimates and confidence intervals (Beaumont et al., 2002).

To overcome these limitations, machine learning methods provide flexible statistical tools that integrate multiple data types, model nonlinear interactions, and quantify variable importance. By combining simulations with supervised learning, researchers can train models to identify factors most strongly associated with genetic variation while controlling for stochastic effects. Similarly, convolutional neural networks (CNNs) allow the statistical extraction of patterns from DNA sequence alignments treated as high-dimensional data, enabling inference without pre-defined summary statistics or population assignments (Flagel et al., 2019).

analysis]Statistical methods in genetic analysis

In quantitative genetics, we often partition phenotypic variance into genetic and environmental components:

$$\sigma_P^2 = \sigma_G^2 + \sigma_E^2$$

$$h^2 = \frac{\sigma_A^2}{\sigma_P^2}$$

$$F_{ST} = \frac{\sigma_{\text{between populations}}^2}{\sigma_{\text{total}}^2}$$

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i (R_i - R)^2$$

$$y = X\beta + Zu + \epsilon$$

$$G_m = \sum_{k=1}^K p_{mk}(1 - p_{mk})$$

$$I_j = \frac{1}{T} \sum_{t=1}^T (L_{\text{perm}}(t) - L_{\text{orig}}(t))$$

$$P(\theta \mid S_{\text{obs}}) \approx P(\theta \mid \| S_{\text{sim}} - S_{\text{obs}} \| < \epsilon)$$

where S_{obs} = observed summary statistics,
 S_{sim} = simulated statistics under parameter θ , and
 ϵ = tolerance threshold. Finally, Convolutional Neural
Networks (CNN), for a single convolution operation is:

$$h_{i,j}^{(l)} = f(\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} w_{m,n}^{(l)} x_{i+m,j+n}^{(l-1)} + b^{(l)})$$

analysis]Supervised machine learning in genetic analysis

Supervised machine learning (ML) is a statistical framework that maps input features (e.g., environmental variables, SNP data) to outcomes (e.g., allelic richness, heterozygosity) based on labeled datasets. Key statistical steps included data preprocessing (normalization, encoding of categorical variables, and missing data imputation), model selection (algorithms such as random forests, gradient boosting machines, or support vector machines), training and validation (k-fold cross-validation or bootstrapping to estimate model performance metrics like RMSE, R²), and inference estimation of feature importance using permutation tests or Gini impurity to identify key drivers of genetic variation). This approach allows quantitative integration of environmental and historical factors, while simulations can expand training datasets to include scenarios not observed in empirical samples, increasing statistical robustness.

Integrating discrete environmental factors within a common analytical framework allows researchers to pinpoint the elements that significantly influence genetic diversity in biological species. This integration could take several forms, such as conducting independent polymorphic genetic analyses and synthesizing the results statistically, or performing a single comprehensive analysis that incorporates environmental factors. One potential approach involves developing a full likelihood framework, though this is a complex and daunting task due to the intricate models required (Beaumont et al., 2002). Alternatively, simulation-based methods that generate genetic data in a spatial context offer a promising direction. Despite the challenges of applying such frameworks to empirical systems; especially those with poorly characterized life parameters like generation length and mutation rates, the integration of simulations within a supervised machine learning framework could streamline the process. Supervised machine learning, a subset of artificial intelligence, involves training a predictive model using pre-classified datasets. In this context, a model would be trained on datasets where only historical or landscape factors are evident. Given that both types of factors usually coexist, pretraining can also utilize data reflecting the presence of both factors. This approach can simplify the development of a framework capable of identifying the factors most influential in shaping genetic diversity in the focal species (Schrider and Kern, 2018). Integrating discrete environmental factors under a supervised machine learning framework offers a robust approach to understanding the processes shaping genetic diversity in biological species. This method simplifies the complex task of model development, providing a practical and efficient tool for researchers to identify key influences on genetic variation. By bridging the gap between historical and landscape genetics, this framework enhances our ability to test hypotheses and interpret genetic data across diverse spatial and temporal scales.

2 Artificial neural network

Integrating discrete environmental factors within a common analytical framework is essential for identifying the influences on genetic diversity in biological species. A promising approach involves the use of convolutional neural networks, a type of artificial neural network that mimics the human brain's neural connections by artificially replicating neuron interactions. Originally developed for image and video classification, convolutional neural networks can be adapted for genetic data analysis by altering data representation methods (Flagel et al., 2019). This simulation-based framework, which employs convolutional neural networks, offers a robust method for analyzing genetic diversity. By adapting convolutional neural networks to process genetic data as images of DNA sequence alignments, researchers can avoid the limitations of traditional summary statistics and a priori decisions. This approach provides a comprehensive tool for identifying the environmental and historical factors that shape genetic diversity, bridging the gap between different genetic analysis disciplines and enhancing our understanding of genetic variation across diverse spatial and temporal scales.

3 Polymorphic genetics

Evolutionary genetics is inherently challenging because each species has a distinct history and responds uniquely to its environment. This uniqueness adds layers of complexity to research, but it also enriches the field by presenting a diverse array of questions to explore and hypotheses to test. The interplay of factors such as life history, habitat specificity, and ecological location shapes genetic patterns in ways that are unique to each species. Taxon-specific traits significantly influence genetic patterns, and trait-based approaches have been instrumental in clarifying the nature of discordant genetic patterns (Sullivan et al., 2019). Many species remain poorly understood in terms of their ecology, evolution, distribution, and population dynamics. This lack of knowledge complicates data analysis, as the choice of analytical methods should ideally be guided by comprehensive information about the focal species. Without such information, researchers risk using inappropriate analytical methods, leading to potentially flawed conclusions. Although numerous programs and frameworks are available to address scientific questions, selecting the right one depends on understanding the species' life history, distribution, and population dynamics. Initially, polymorphic genetics relied on various types of genetic data, including sequence data based on gene genealogies. More recently, researchers have used summary statistics, such as nucleotide diversity, calculated from discrete units (Garrick et al., 2020). In contrast, earlier studies based on allelic data, such as microsatellites, often analyzed metrics

among demes or individuals. The advent of Single-Nucleotide Polymorphism data has revolutionized genetic research, as these data are economical to generate, even in non-model systems. However, a significant distinction remains in how genetic variation is partitioned, whether by populations or individuals. This partitioning can impact the interpretation of genetic patterns and the subsequent conclusions drawn from the data.

The unique genetic patterns observed among taxonomically similar species underscore the importance of taxon-specific traits. Understanding these traits helps elucidate the reasons behind discordant genetic patterns and provides insights into the evolutionary processes at play. Knowledge gaps about a species' ecology, evolution, distribution, and population dynamics have significant implications for the analysis of genetic data. Accurate data interpretation requires an informed selection of analytical methods, which in turn depends on comprehensive species-specific information. Evolutionary genetics is a complex field due to the unique histories and environmental responses of each species. This complexity, while challenging, also makes research more intriguing by offering a wide array of questions and hypotheses. The development and application of appropriate analytical methods depend on a thorough understanding of the species under study. Advances in genetic data collection, such as the use of Single-Nucleotide Polymorphism, have improved our ability to study genetic variation, but challenges remain in accurately partitioning and interpreting this variation. By focusing on taxon-specific traits and leveraging new technologies, researchers can continue to uncover the intricate details of evolutionary genetics and enhance our understanding of biodiversity.

computation|Approximate Bayesian
computation

The complexity of evolutionary genetics requires advanced methods for accurate data analysis. While traditional approaches like approximate Bayesian computation rely heavily on researcher choices in summarizing genetic variation, convolutional neural networks offer a promising alternative. By using alignment images, convolutional neural networks can enhance the capacity to infer both discrete and continuous genetic processes without the biases inherent in traditional methods. The approximate Bayesian computation allows researchers to make statistical inferences based on simulated data that approximate the observed data. However, the effectiveness of approximate Bayesian computation depends on how researchers summarize genetic variation. Poor summarization can reduce the statistical power of the analysis and lead to biased results. This is especially problematic in phylogeography and landscape genetics, where model-based methods have become prevalent over the last decades (Zamudio et al., 2016). The choice of analytical model often reflects the researchers' beliefs about the processes shaping genetic variation. Since models are inherently simplified

representations of reality, they can lead to biased or erroneous inferences, particularly if important factors like genetic drift, selection, hybridization, and biological interactions are overlooked. Ignoring environmental-level analyses can result in confirmation bias, where findings align with preconceived notions rather than empirical evidence. convolutional neural networks provide a robust alternative to approximate Bayesian computation by processing genetic data as images of DNA sequence alignments. This approach retains the essential features of genetic diversity and does not require a priori decisions about population structure or summary statistics. The convolutional neural networks can capture both discrete and continuous genetic processes, offering a more comprehensive and unbiased analysis framework. The integration of convolutional neural networks into genetic data analysis offers a robust solution to the limitations of traditional methods like approximate Bayesian computation. By processing genetic data as images, convolutional neural networks can capture the complexity of genetic variation without the biases introduced by prior assumptions. This approach enhances the capacity to infer discrete and continuous genetic processes, providing a more accurate and comprehensive understanding of the factors shaping genetic diversity. As the field of evolutionary genetics continues to evolve, leveraging advanced techniques like convolutional neural networks will be crucial for uncovering the intricate details of genetic variation and evolutionary processes.

4 Conclusion

Integrating advanced statistical frameworks; supervised machine learning, CNNs, and simulation-based inference, into genetic analyses enhances our ability to quantify the drivers of genetic diversity. These approaches enable robust, scalable, and interpretable inference across complex landscapes and evolutionary histories. By combining traditional population genetics theory with modern statistical learning methods, researchers can obtain actionable insights for conservation, biodiversity management, and understanding evolutionary processes.

5 References

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